

FISCAL AND MONETARY POLICY INTERACTIONS

FISCAL AND MONETARY POLICY 2026

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June 1, 2026

ROADMAP

- **Fixed-regime benchmark**: root counts and shock responses.
- **Regime switching**: how future regimes change today's coefficients.
- **Identification**: why equilibrium correlations do not reveal structural regimes.
- **Valuation**: why low rates and default require extra structure.
- **Fiscal compatibility**: how policy rules restrict the target price path.

HOW THE PIECES FIT

We start from one benchmark and then change one ingredient at a time.

- The first model is the **linear Fisher–budget system**; it gives root counts and shock coefficients.
- Switching keeps the same logic but lets policy coefficients follow a **Markov chain**.
- Identification asks what an econometrician sees from **equilibrium data**.
- Low rates and default change the **valuation equation**.
- Fiscal compatibility asks whether a target price path has enough **fiscal backing**.

FIXED-REGIME BENCHMARK: ROOT-COUNT DETERMINACY

	Fiscal passive	Fiscal active
Monetary active	Regime M	AM/AF: no eq.
Monetary passive	PM/PF: indeterminate	Regime F

- One active + one passive $\Rightarrow$ determinate equilibrium.
- Other combinations yield either no bounded equilibrium or indeterminacy.
- These statements assume the regime is fixed forever and default is ruled out.

THE FIXED-REGIME SYSTEM

Use **level deviations of gross nominal objects**, not log deviations:

$$i_t \equiv R_t - \bar{R}, \quad \pi_t \equiv \Pi_t - \bar{\Pi}, \quad \bar{\Pi} = 1, \quad \bar{R} = \beta^{-1}.$$

Linearizing Fisher gives $\mathbb{E}_t \pi_{t+1} = \beta i_t$. The policy rules and normalized budget constraint are

$$i_t = \phi_\pi \pi_t + \theta_t,$$

$$\tau_t = \gamma b_{t-1} + \psi_t,$$

$$b_t = \beta^{-1} b_{t-1} - \beta^{-1} \pi_t - \tau_t.$$

- b_t, τ_t are normalized so the inflation-revaluation coefficient is β^{-1} .
- θ_t is a monetary-rule intercept shock; $\psi_t > 0$ is a surplus shock.
- Monetary active means **$\beta \phi_\pi > 1$** .

TIMING OF THE DEBT STATE

The predetermined state in period t is beginning-of-period inherited debt:

$$b_{t-1}.$$

Within period t :

- π_t revalues inherited nominal liabilities.
- τ_t is the period- t primary surplus.
- The government budget produces b_t , the end-of-period debt stock.

Therefore

$$\pi_t = hb_{t-1} + g^\theta \theta_t + g^\psi \psi_t, \quad b_t \text{ is next period's state.}$$

MATRIX FORM

Combining the policy rules with Fisher and the government budget constraint:

$$\begin{bmatrix} \mathbb{E}_t \pi_{t+1} \\ b_t \end{bmatrix} = \underbrace{\begin{bmatrix} \phi_\pi \beta & 0 \\ -\beta^{-1} & \beta^{-1} - \gamma \end{bmatrix}}_A \begin{bmatrix} \pi_t \\ b_{t-1} \end{bmatrix} + \begin{bmatrix} \beta \theta_t \\ -\psi_t \end{bmatrix}.$$

- π_t is a **jump** variable.
- b_{t-1} is a **predetermined** state.
- Blanchard–Kahn conditions require exactly **one unstable root**.

EIGENVALUE ALGEBRA

The matrix is triangular, so:

$$\det(A - \lambda I) = (\phi_{\pi}\beta - \lambda)(\beta^{-1} - \gamma - \lambda),$$

$$\lambda_{\pi} = \phi_{\pi}\beta,$$

$$\lambda_b = \beta^{-1} - \gamma.$$

- Monetary active: $\lambda_{\pi} > 1 \Leftrightarrow \phi_{\pi} > 1/\beta$.
- Monetary passive: $\lambda_{\pi} < 1$.
- Fiscal passive: $|\lambda_b| < 1$.
- Fiscal active (positive-root case): $\lambda_b > 1 \Leftrightarrow \gamma < \beta^{-1} - 1$.
- Explosive negative-root fiscal cases are excluded from the simple taxonomy below.

REGIMES ARE ROOT COUNTS

Regime	Roots	Unstable roots	Outcome
AM/PF	$\lambda_\pi > 1, \lambda_b < 1$	one	Regime M
PM/AF	$\lambda_\pi < 1, \lambda_b > 1$	one	Regime F
AM/AF	$\lambda_\pi > 1, \lambda_b > 1$	two	No bounded eq.
PM/PF	$\lambda_\pi < 1, \lambda_b < 1$	zero	Indeterminate

- **Regime M**: boundedness selects inflation; fiscal policy stabilizes debt.
- **Regime F**: the price level jumps to put inherited real debt on the fiscal valuation path.
- Root count is a **local linear result** (default-free, fixed rules), not a full historical classification.

FIXED-REGIME SOLUTION: SHOCK PROCESSES

Define

$$\lambda_{\pi} \equiv \phi_{\pi} \beta, \quad \lambda_b \equiv \beta^{-1} - \gamma.$$

Let policy shocks follow

$$\theta_t = \rho_{\theta} \theta_{t-1} + \varepsilon_t^{\theta}, \quad \psi_t = \rho_{\psi} \psi_{t-1} + \varepsilon_t^{\psi}, \quad |\rho_{\theta}|, |\rho_{\psi}| < 1.$$

Guess a bounded linear solution:

$$\pi_t = h b_{t-1} + g^{\theta} \theta_t + g^{\psi} \psi_t.$$

FIXED-REGIME SOLUTION: DEBT LAW

The debt law implied by the guess is

$$b_t = (\lambda_b - \beta^{-1}h)b_{t-1} - \beta^{-1}g^\theta\theta_t - (1 + \beta^{-1}g^\psi)\psi_t.$$

The coefficient on inherited debt is

$$\lambda_b - \beta^{-1}h.$$

- In **Regime M** this root must be stable directly.
- In **Regime F** the price level jumps so this root becomes the stable monetary root.

FIXED-REGIME SOLUTION: COEFFICIENT EQUATIONS

Matching coefficients in the Fisher equation gives

$$\begin{aligned}h(\lambda_b - \beta^{-1}h) &= \lambda_\pi h, \\g^\theta(\rho_\theta - \lambda_\pi - \beta^{-1}h) &= \beta, \\g^\psi(\rho_\psi - \lambda_\pi - \beta^{-1}h) &= h.\end{aligned}$$

- The first equation has two roots:

$$h = 0 \quad \text{or} \quad h = \beta(\lambda_b - \lambda_\pi).$$

- **Boundedness** selects which root is the equilibrium root.

REGIME M SOLUTION: COEFFICIENTS

In Regime M,

$$\lambda_\pi > 1, \quad |\lambda_b| < 1.$$

Bounded inflation selects

$$h_M = 0, \quad g_M^\theta = -\frac{\beta}{\lambda_\pi - \rho_\theta}, \quad \boxed{g_M^\psi = 0}.$$

- The zero fiscal-shock coefficient is the algebraic insulation result.

REGIME M SOLUTION: RESPONSES

The AM/PF solution is

$$\pi_t = -\frac{\beta}{\lambda_\pi - \rho_\theta} \theta_t,$$

and

$$b_t = \lambda_b b_{t-1} + \frac{1}{\lambda_\pi - \rho_\theta} \theta_t - \psi_t.$$

The nominal-rate response is

$$i_t = \phi_\pi \pi_t + \theta_t = -\frac{\rho_\theta}{\lambda_\pi - \rho_\theta} \theta_t.$$

The coefficient on θ_t is negative for $\rho_\theta > 0$, so a rate-increase IRF sets $\theta_0 < 0$.

REGIME M: FISCAL SHOCKS ARE INSULATED

Set $\theta_t = 0$ and consider a **fiscal shock** ψ_t :

$$\pi_t = 0, \quad i_t = 0, \quad b_t = \lambda_b b_{t-1} - \psi_t.$$

Since fiscal policy is passive,

$$|\lambda_b| < 1,$$

so debt returns to its steady state after the shock.

- A tax cut is $-\psi_t$: it raises **debt**, not **inflation**.
- **Active monetary policy** pins down inflation; **passive fiscal policy** absorbs the fiscal shock through future surpluses.
- This is the fixed-regime benchmark that **switching will overturn**.

REGIME F SOLUTION: PM/AF

In Regime F,

$$\lambda_{\pi} < 1, \quad \lambda_b > 1.$$

Bounded debt selects

$$h_F = \beta(\lambda_b - \lambda_{\pi}).$$

The shock coefficients are

$$g_F^{\theta} = -\frac{\beta}{\lambda_b - \rho_{\theta}}, \quad g_F^{\psi} = -\frac{\beta(\lambda_b - \lambda_{\pi})}{\lambda_b - \rho_{\psi}}.$$

REGIME F: FISCAL SHOCKS MOVE INFLATION

The **Regime F** solution is

$$\pi_t = \beta(\lambda_b - \lambda_\pi)b_{t-1} - \frac{\beta}{\lambda_b - \rho_\theta}\theta_t - \frac{\beta(\lambda_b - \lambda_\pi)}{\lambda_b - \rho_\psi}\psi_t.$$

Debt evolves according to

$$b_t = \lambda_\pi b_{t-1} + \frac{1}{\lambda_b - \rho_\theta}\theta_t - \frac{\lambda_\pi - \rho_\psi}{\lambda_b - \rho_\psi}\psi_t.$$

- The **fiscal shock coefficient** in inflation is nonzero.
- **Active fiscal policy** makes the price level revalue inherited nominal debt.

REGIME F: MONETARY INTERCEPT SHOCKS

Set $\psi_t = 0$ and take constant surpluses, $\gamma = 0$, so $\lambda_b = \beta^{-1}$.

With inherited debt at steady state, $b_{t-1} = 0$, the impact response is

$$\pi_t = -\frac{\beta}{\beta^{-1} - \rho_\theta} \theta_t.$$

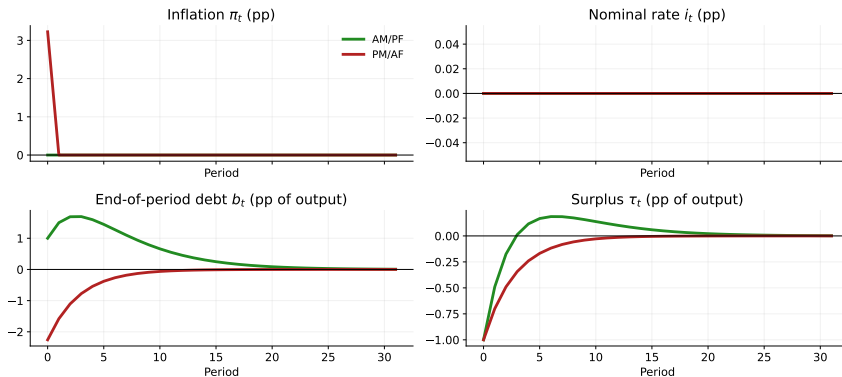
The observed nominal-rate response is

$$i_t = \left(1 - \frac{\lambda_\pi}{\beta^{-1} - \rho_\theta}\right) \theta_t.$$

- A positive intercept shock lowers inflation on impact.
- With $\phi_\pi(F) = 0$, $i_t = \theta_t$, so the plotted $\theta_0 = -1$ is a rate cut in PM/AF.

IRF: FISCAL SHOCK WITH FIXED REGIMES

Fixed regimes, no switching: fiscal tax-cut shock



Units: π_t, i_t in percentage points; b_t is end-of-period debt; b_t, τ_t in points of output. Shock: $\psi_0 = -1$ pp of output.

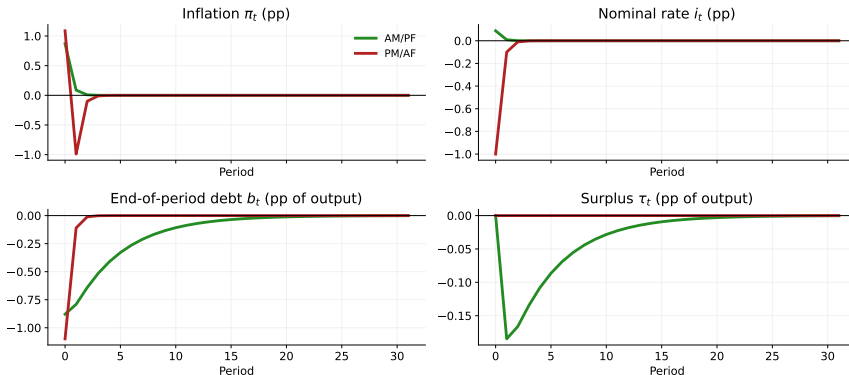
Calib: $\beta = 0.99, \rho_\theta = 0.10, \rho_\varphi = 0.70$; M $\phi_\pi = 1.25, \lambda_\theta = 0.80$; F $\phi_\pi = 0.00, \gamma = 0.00$.

No switching: AM/PF and PM/AF are fixed forever. Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock:

$$\psi_0 = -1.$$

IRF: MONETARY INTERCEPT SHOCK WITH FIXED REGIMES

Fixed regimes, no switching: monetary-rule intercept shock



Units: π_t, i_t in percentage points; b_t, τ_t in points of output. Shock: $\theta_0 = -1$.

Calib: $\beta = 0.99, \rho_\theta = 0.10, \rho_\gamma = 0.70$; M $\phi_\pi = 1.25, \lambda_0 = 0.80$; F $\phi_\pi = 0.00, \gamma = 0.00$.

No switching: AM/PF and PM/AF are fixed forever. Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock:

$$\theta_0 = -1.$$

MOTIVATION: POLICY-RULE COEFFICIENTS ARE TIME-VARYING

- The fixed-regime solution assumed ϕ_π and γ are constant forever.
- 1942–1951: Fed pegged 3-month Treasury bills at 3/8% to support war finance.
- 1965–1979: empirical Taylor coefficient fails the Taylor principle.
- 1979–2007: Volcker disinflation and pre-GFC era. Active monetary, mostly passive fiscal.
- 2008–2015, 2020–2022: ZLB / near-ZLB episodes; debt/GDP rises sharply.
- 2022–2026: high-rate anti-inflation regime; fiscal regime still disputed.

WHAT CHANGES AFTER THE FIXED BENCHMARK?

The fixed-regime solution gave two clean results:

$$g_M^\psi = 0, \quad g_F^\psi \neq 0.$$

The rest of the lecture asks what survives when we relax one assumption at a time.

- **Regime switching**: policy coefficients are stochastic, so today's coefficients depend on transition probabilities.
- **Identification**: equilibrium correlations mix policy rules with valuation equations.
- **Low rates and default**: the simple valuation equation needs extra structure.
- **Fiscal compatibility**: an inflation target requires fiscal backing of the target price path.

WHY FIXED REGIMES ARE INSUFFICIENT

- The root-count result is conditional on policy coefficients being fixed forever.
- Historical policy rules are **time-varying**: wartime pegs, Volcker disinflation, ZLB episodes, and post-2022 tightening reflect different monetary–fiscal environments.
- Technical question: what replaces the single matrix A when the policy-rule coefficients follow a **Markov chain**?
- Answer: solve for **regime-contingent coefficients** linked by the transition matrix.

REGIME SWITCHING

MARKOV-SWITCHING POLICY RULES

$$i_t = \bar{i}(\sigma_t) + \phi_\pi(\sigma_t) \pi_t + \theta_t,$$

$$\tau_t = \gamma_0(\sigma_t) + \gamma_1(\sigma_t) b_{t-1} + \psi_t,$$

where $\sigma_t \in \{M, F\}$ follows

$$\Pr(\sigma_{t+1} = j \mid \sigma_t = i) = p_{ij}, \quad \mathcal{P} = \begin{bmatrix} p_{MM} & p_{MF} \\ p_{FM} & p_{FF} \end{bmatrix}.$$

- **Regime M**: AM/PF, $\phi_\pi(M) > 1/\beta$ and $|\beta^{-1} - \gamma_1(M)| < 1$.
- **Regime F**: PM/AF, $\phi_\pi(F) < 1/\beta$ and $\beta^{-1} - \gamma_1(F) > 1$ in the positive-root case.
- Private agents observe σ_t and know $\mathcal{P}$.
- Below write $\phi_{\pi,i} \equiv \phi_\pi(i)$ and $\gamma_i \equiv \gamma_1(i)$.

REGIME-CONTINGENT SOLUTION FORM

Assume a bounded linear solution:

$$\pi_t = h_{\sigma_t} b_{t-1} + g_{\sigma_t}^{\theta} \theta_t + g_{\sigma_t}^{\psi} \psi_t.$$

The expectation in the Fisher equation is now a cross-regime object:

$$\mathbb{E}_t \pi_{t+1} = \sum_{j \in \{M, F\}} p_{\sigma_t j} \mathbb{E}_t \left[h_j b_t + g_j^{\theta} \theta_{t+1} + g_j^{\psi} \psi_{t+1} \right].$$

- The coefficients $(h_j, g_j^{\theta}, g_j^{\psi})$ are solved jointly across regimes.
- Today's inflation response depends on $\mathcal{P}$, not only on today's policy coefficients.

SWITCHING ALGEBRA: DEBT LAW OF MOTION

Suppress intercepts and shocks first. In state $i \in \{M, F\}$, conjecture

$$\pi_t = h_i b_{t-1}.$$

The government budget equation implies

$$b_t = \left(\beta^{-1} - \gamma_i - \beta^{-1} h_i \right) b_{t-1} \equiv q_i(h_i) b_{t-1}.$$

Thus $q_i(h_i)$ is the regime- i debt-transition coefficient:

$$q_i(h_i) = \beta^{-1} - \gamma_i - \beta^{-1} h_i.$$

SWITCHING ALGEBRA: DEBT COEFFICIENT EQUATION

The Fisher equation gives

$$\beta \phi_{\pi,i} h_i b_{t-1} = \sum_{j \in \{M,F\}} p_{ij} h_j b_t = \left(\sum_j p_{ij} h_j \right) q_i(h_i) b_{t-1}.$$

Therefore the two regime coefficients solve

$$\boxed{\beta \phi_{\pi,i} h_i = \left(\sum_j p_{ij} h_j \right) \left(\beta^{-1} - \gamma_i - \beta^{-1} h_i \right)}, \quad i = M, F.$$

SWITCHING ALGEBRA: FISCAL-SHOCK SETUP

Let $H_i \equiv \sum_j p_{ij} h_j$ and $G_i^\psi \equiv \sum_j p_{ij} g_j^\psi$, with persistent surplus shock $\psi_{t+1} = \rho_\psi \psi_t + \varepsilon_{t+1}^\psi$.
Conjecture

$$\pi_t = h_i b_{t-1} + g_i^\psi \psi_t.$$

The coefficient on ψ_t in next period's debt is

$$\frac{\partial b_t}{\partial \psi_t} = - \left(1 + \beta^{-1} g_i^\psi \right).$$

Matching ψ_t coefficients in Fisher (using $\mathbb{E}_t \psi_{t+1} = \rho_\psi \psi_t$):

$$\beta \phi_{\pi, i} g_i^\psi = -H_i \left(1 + \beta^{-1} g_i^\psi \right) + \rho_\psi G_i^\psi, \quad i = M, F.$$

SWITCHING ALGEBRA: FISCAL-SHOCK RESULT

Collecting terms, the regime coefficients solve the coupled system

$$\left(\beta \phi_{\pi,i} + \beta^{-1} H_i \right) g_i^\psi - \rho_\psi \sum_j p_{ij} g_j^\psi = -H_i, \quad i = M, F.$$

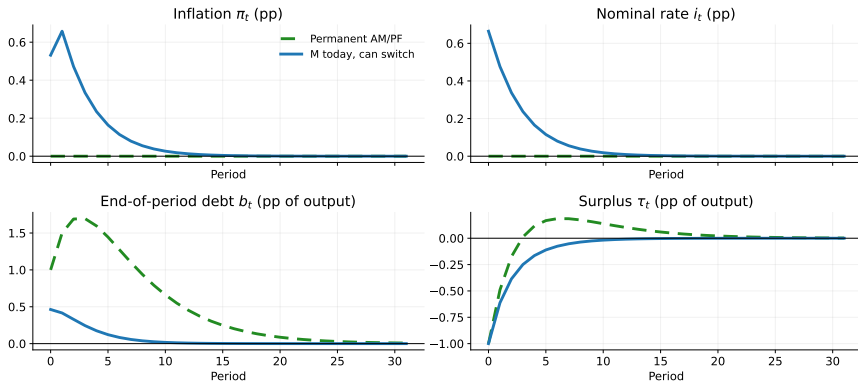
The i.i.d. case $\rho_\psi = 0$ gives the closed form

$$g_i^\psi = -\frac{H_i}{\beta \phi_{\pi,i} + \beta^{-1} H_i}.$$

- Permanent AM/PF: $p_{MF} = 0$ and $h_M = 0$ imply $H_M = 0$ and $g_M^\psi = 0$.
- Switching with $p_{MF} > 0$ and $h_F \neq 0$ generically makes $H_M \neq 0$, so $g_M^\psi \neq 0$.
- A tax cut is $-\psi_t$, so the inflation response has the opposite sign of g_i^ψ .

IRF: FISCAL SHOCK WITH SWITCHING RISK

Regime M today with $p(M \text{ to } F)=0.10$: fiscal tax-cut shock

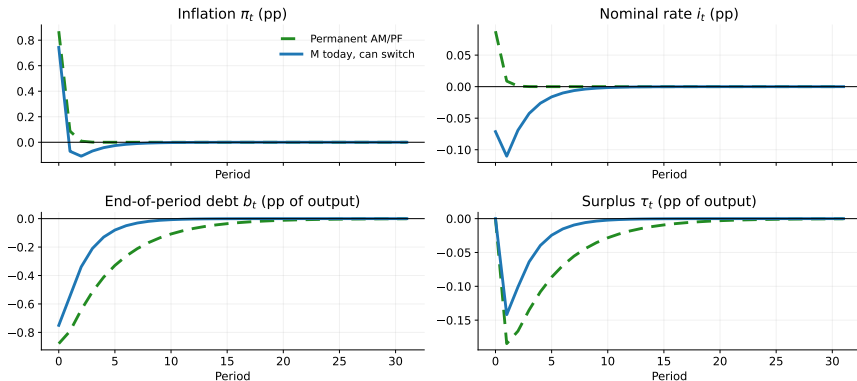


Units: π_t, i_t in percentage points; b_t, τ_t in points of output. Shock: $\psi_0 = -1$ pp of output.
Calib: $\beta = 0.99, \rho_\theta = 0.10, \rho_\psi = 0.70$; M $\phi_\pi = 1.25, \lambda_\theta = 0.80$; F $\phi_\pi = 0.00, \gamma = 0.00$; $p_{MF} = 0.10, p_{FF} = 0.85$.

Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\psi_0 = -1$. Calibration: $\beta = .99, \rho_\theta = .10, \rho_\psi = .70,$
 $p_{MF} = .10, p_{FF} = .85$.

IRF: MONETARY RATE-INCREASE SHOCK WITH SWITCHING RISK

Regime M today with $p(M \text{ to } F)=0.10$: monetary-rule intercept shock



Units: π_t, i_t in percentage points; b_t is end-of-period debt; b_t, τ_t in points of output. Shock: $\theta_0 = -1$.
 Calib: $\beta = 0.99, \rho_\theta = 0.10, \rho_\psi = 0.70$; M $\phi_\pi = 1.25, \lambda_\theta = 0.80$; F $\phi_\pi = 0.00, \gamma = 0.00$; $p_{MF} = 0.10, p_{FF} = 0.85$.

Units: π_t, i_t in pp; b_t, τ_t in pp of output. Shock: $\theta_0 = -1$. Calibration: $\beta = .99, \rho_\theta = .10, \rho_\psi = .70,$
 $p_{MF} = .10, p_{FF} = .85$.

TAYLOR COEFFICIENTS CHANGE VALUATION COEFFICIENTS

In the i.i.d. case, a stronger Taylor response changes the equilibrium valuation mapping:

$$g_i^\psi = -\frac{H_i}{\beta\phi_{\pi,i} + \beta^{-1}H_i}, \quad H_i \equiv \sum_j p_{ij}h_j.$$

If agents assign positive probability to **active fiscal policy**, the current fiscal-shock coefficient depends on both today's monetary feedback and future Regime-F valuation terms;

$\partial g_M^\psi / \partial \phi_\pi(M)$ can be positive.

- The mechanism is **coefficient selection across regimes**, not an IS-curve demand channel.
- Active monetary policy today need not insulate inflation from fiscal shocks when future fiscal backing may be active.

IDENTIFICATION: REGIMES ARE NOT OBSERVED DIRECTLY

- Cochrane (1999), Leeper–Walker (2013): in simple linear settings, the same observables can often be rationalized under either regime.
- Reason: with suitable policy-rule and shock specifications, the same state vector

$$X_t \equiv (\epsilon_t^M, \epsilon_t^F, \hat{b}_{t-1})$$

can support different structural rules.

- The same equilibrium mapping $i_t = f(X_t)$ may be generated by different policy rules because structural rules restrict **off-equilibrium** behavior, not only equilibrium correlations.
- Simple correlations **cannot** distinguish the regimes.

TAYLOR-RULE OLS TRAP: WHAT OLS ESTIMATES

- **Structural rule** and monetary shock:

$$i_t = \phi\pi\pi_t + \theta_t, \quad \theta_t = \rho_\theta\theta_{t-1} + \varepsilon_t^\theta.$$

- **Equilibrium data** obey:

$$i_t = f(\theta_t), \quad \pi_t = g(\theta_t).$$

- OLS estimates

$$\hat{\delta} = \frac{\text{Cov}(i_t, \pi_t)}{\text{Var}(\pi_t)},$$

not the structural Taylor coefficient $\phi\pi$.

CORRECT IDENTIFICATION STRATEGY

- Impose the bond valuation equation as a cross-equation restriction.
- Estimate monetary and fiscal rules **jointly** (Chung–Leeper 2007).
- Use a fully specified DSGE model with both regimes (Traum–Yang; Leeper–Traum–Walker).
- Allow Markov-switching coefficients (Bianchi; Bianchi–Ilut; Chen–Leeper–Leith).
- Reduced-form correlations do not suffice on their own.

VALUATION WITH LOW RATES AND DEFAULT

BASELINE FTPL VALUATION WITH AN SDF

Maintained assumption: promised nominal payments are always honored.

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j} \tau_{t+j}.$$

- Left side: **real value** of inherited nominal promises.
- Right side: **risk-adjusted present value** of primary surpluses.
- If the operator converges, debt is default-free, and surpluses do not passively validate every P_t , the equation can **select** P_t .
- The next two modifications ask what happens when **discounting is weak** or **repayment is defaultable**.

TWO WAYS THE VALUATION EQUATION CAN FAIL

Baseline default-free valuation:

$$\frac{R_{t-1}B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j} \tau_{t+j}.$$

- **Low rates:** $\mathbb{E}_t \sum_{j \geq 0} M_{t,t+j} \tau_{t+j}$ may not give a unique finite valuation.
- **Default:** $R_{t-1}B_{t-1}/P_t$ is not what bondholders actually receive.

With default,

$$\rho_t \frac{R_{t-1}B_{t-1}}{P_t} = \text{PV}_t(\tau), \quad \rho_t \equiv 1 - \delta_t.$$

Low rates are a discounting/uniqueness problem. Default is a payoff/adjustment-margin problem.

BASSETTO–CUI: DEBT LAW

Use compact debt-output notation:

$$b_t^Y \equiv \frac{B_t}{P_t Y_t}, \quad x_t \equiv \frac{\tau_t}{Y_t}, \quad m_{t+1} \equiv \frac{1 + r_{t+1}}{1 + g_{t+1}}.$$

The government budget is

$$b_{t+1}^Y = m_{t+1}(b_t^Y - x_t),$$

where τ_t is the real primary surplus and x_t is the surplus–output ratio.

- Low government-debt returns mean $m_{t+1} < 1$ for sustained periods.
- Bassetto–Cui ask whether the fiscal valuation equation still **selects the price level**.

WHY $r < g$ WEAKENS PRICE-LEVEL SELECTION

For simplicity, take a constant return-growth factor $m < 1$:

$$b_{t+1}^Y = m(b_t^Y - x_t).$$

If primary surpluses are zero,

$$x_t = 0 \quad \Rightarrow \quad b_t^Y = m^t b_0^Y \rightarrow 0.$$

So every initial real debt value $b_0^Y = B_0/(P_0 Y_0)$ is dynamically admissible.

But B_0 and Y_0 are given, so different admissible b_0^Y values imply different admissible P_0 values.

$$r < g \quad \Rightarrow \quad \text{boundedness alone may not select } P_0.$$

To keep b_t^Y positive and bounded away from zero when $m < 1$, average primary surpluses must be negative.

BASSETTO–CUI: THE REASON FOR LOW RATES MATTERS

Mechanism	Why government returns are low	FTPL implication
Risk adjustment	Debt is valuable in high-SDF states, or backing is high in high-SDF states	Risk-adjusted PV can still converge; FTPL can survive.
Dynamic inefficiency	Debt is partly a store-of-value bubble	Multiple price levels can be consistent with equilibrium.
Liquidity services	Debt carries a convenience yield	Uniqueness depends on debt demand and surplus path; low-return uniqueness is not generic.

$$r - g < 0 \not\Rightarrow \text{FTPL failure.}$$

The relevant object is the risk-adjusted valuation operator and the asset-demand mechanism.

LOW-RATE TAKEAWAY

- Low returns are not one case; their fiscal-theory implications depend on **why** returns are low.
- FTPL selection is most robust in the **risk-premium** case.
- Selection can fail under **dynamic inefficiency**.
- With **liquidity services**, uniqueness depends on debt demand and the surplus path.
- Persistent low returns can coexist with stable or rising positive debt only if primary deficits occur on average.
- Even where uniqueness fails, the fiscal valuation equation can still provide a **lower bound** for the price level.

DEFAULTABLE DEBT PRICES REPAYMENT, NOT PROMISES

Let δ_t be Uribe's default/subsidy rate on inherited nominal liabilities, and define

$$\rho_t \equiv 1 - \delta_t.$$

Then

$$\rho_t \frac{R_{t-1} B_{t-1}}{P_t} = \text{PV}_t(\tau).$$

The valuation equation pins down real repayment,

$$\frac{\rho_t}{P_t},$$

not separately the price level P_t and repayment rate ρ_t .

Convention: in Uribe's benchmark δ_t is not constrained to be nonnegative; a negative δ_t is a subsidy to bondholders.

BOND PRICING WITH DEFAULT RISK

In this block, R_t^d is the promised return on defaultable public debt. The default-free nominal rate is R_t^f . The one-period nominal bond satisfies

$$1 = \mathbb{E}_t \left[M_{t,t+1} R_t^d \frac{P_t}{P_{t+1}} \right].$$

- With constant discounting, $M_{t,t+1} = \beta$, giving Uribe's simplified expression.
- **Default risk** operates inside the FTPL logic, through the asset-pricing problem.
- Monetary policy affects whether fiscal stress is **forecastable** and therefore priced at issuance.

URIBE: MONETARY TIMING CHOOSES THE ADJUSTMENT MARGIN

One-period nominal debt, active fiscal policy, fiscal-news shocks.

Policy rule	What is anchored?	Fiscal-news adjustment
Current-looking Taylor rule	Current inflation, by boundedness	Surprise default/subsidy under price stability.
Forward-looking Taylor rule	Expected inverse inflation	Realized price level absorbs fiscal news; no default in benchmark.
Price-level peg	The price level itself	Default absorbs fiscal news; spreads become forecastable.

Small timing differences change the equilibrium distribution of default.

CURRENT-LOOKING RULE: PRICE STABILITY LEAVES DEFAULT AS THE MARGIN

Let $\Pi_t \equiv P_t/P_{t-1}$ and

$$R_t^d = R^* + \alpha(\Pi_t - \Pi^*), \quad R^* = \frac{\Pi^*}{\beta}.$$

If default is ruled out, bond pricing gives

$$\Pi_{t+1} - \Pi^* = \alpha\beta(\Pi_t - \Pi^*).$$

If $\alpha\beta > 1$, boundedness selects $\Pi_t = \Pi^*$, so fiscal news cannot be absorbed by persistent inflation deviations. The no-default condition is

$$\frac{R_{t-1}^d B_{t-1}}{P_t} = \text{PV}_t(\tau).$$

If fiscal news breaks this equality, the model needs one of three margins: P_t , δ_t , or future τ_t . Under strict price stability and active fiscal policy, δ_t moves.

CURRENT-LOOKING RULE: DEFAULT IS UNFORECASTABLE IN URIBE'S CONVENTION

If $\Pi_t = \Pi^*$ and

$$R_t^d = \frac{\Pi^*}{\beta},$$

bond pricing becomes

$$1 = \mathbb{E}_t(1 - \delta_{t+1}).$$

Therefore

$$\mathbb{E}_t \delta_{t+1} = 0.$$

Realized δ_{t+1} can still move: under Uribe's unconstrained default/subsidy convention, bad fiscal news produces default and good fiscal news produces bondholder subsidies, which cancel in expectation.

REALIZED DEFAULT IS FISCAL NEWS RELATIVE TO ISSUANCE EXPECTATIONS

Define the date- t -forward fiscal backing expected at date q :

$$PV_{t|q} \equiv \sum_{h=0}^{\infty} \beta^h \mathbb{E}_q \tau_{t+h}.$$

Under the target path and one-period debt,

$$\delta_t = 1 - \frac{PV_{t|t}}{PV_{t|t-1}}, \quad t \geq 1.$$

The denominator is last period's expectation of the surplus stream starting at date t .

Bad news: $PV_{t|t} < PV_{t|t-1} \Rightarrow \delta_t > 0$, good news: $PV_{t|t} > PV_{t|t-1} \Rightarrow \delta_t < 0$.

FORWARD-LOOKING RULE: EXPECTED INFLATION IS ANCHORED, REALIZATIONS CAN MOVE

Uribe's forward-looking rule reacts to expected inverse inflation:

$$R_t^d = R^* + \alpha \left(\frac{1}{\mathbb{E}_t(P_t/P_{t+1})} - \Pi^* \right).$$

With $\delta_t = 0$, there is an equilibrium with

$$R_t^d = \frac{\Pi^*}{\beta}, \quad \frac{1}{\mathbb{E}_t(P_t/P_{t+1})} = \Pi^*.$$

But the valuation equation still requires

$$\frac{R_{t-1}^d B_{t-1}}{P_t} = \text{PV}_t(\tau).$$

Fiscal news moves the realized price level through $\text{PV}_t(\tau)$.

The rule anchors expected inverse inflation while leaving each realized rate free to move.

PRICE-LEVEL PEG: DEFAULT BECOMES FORECASTABLE

Under a price-level peg, $P_t = 1$. Fiscal news cannot be absorbed by inflation.

$$\delta_t = 1 - \frac{PV_t(\tau)}{R_{t-1}^d B_{t-1}}.$$

Bond pricing under the peg gives

$$1 = \beta R_t^d \mathbb{E}_t(1 - \delta_{t+1}), \quad R_t^f = \beta^{-1}.$$

Therefore

$$\frac{R_t^d}{R_t^f} = \frac{1}{\mathbb{E}_t(1 - \delta_{t+1})}.$$

Under constant discounting, persistent fiscal weakness lowers expected repayment and raises the sovereign spread.

DELAYED DEFAULT: INFLATION PATH

Perfect-foresight transition. Assume

$$\delta_t = 0 \quad \text{for } 0 \leq t < T.$$

Under the current-looking Taylor rule,

$$\Pi_t = \Pi^* + (\alpha\beta)^t(\Pi_0 - \Pi^*).$$

If

$$\alpha\beta > 1 \quad \text{and} \quad \Pi_0 > \Pi^*,$$

the inflation gap grows while default is delayed.

Temporary delay does not remove the fiscal imbalance.

DELAYED DEFAULT: TERMINAL ADJUSTMENT

Let the no-default continuation inflation at date T be

$$\Pi_T^{nd} = \Pi^* + (\alpha\beta)^T (\Pi_0 - \Pi^*).$$

At date T , default restores the target inflation path. The required default/subsidy rate is

$$\delta_T = 1 - \frac{\Pi^*}{\Pi_T^{nd}} = 1 - \frac{\Pi^*}{\Pi^* + (\alpha\beta)^T (\Pi_0 - \Pi^*)}.$$

Longer delay raises the required terminal adjustment.

In Uribe's transition, real public debt can remain flat until the crisis.

FISCAL COMPATIBILITY

FISCAL COMPATIBILITY OF INFLATION TARGETING

Given **inherited nominal liabilities** and target price path P_t^* :

$$\frac{R_{t-1}B_{t-1}}{P_t^*} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j} \tau_{t+j}.$$

- Exact inflation targeting imposes a **restriction on fiscal backing**.
- **Too-low fiscal backing** $\Rightarrow$ higher P_t , default, or future fiscal correction.
- **Too-high fiscal backing** $\Rightarrow$ lower P_t , requiring fiscal relaxation or other valuation adjustment.
- Inflation targeting is a **joint monetary–fiscal commitment**.

FISCAL RULES AS BACKING RESTRICTIONS

- Sweden: net lending target is 1/3% of GDP over the cycle through 2026; balance target from 2027.
- Switzerland: debt brake caps spending at cyclically adjusted receipts.
- Germany: debt brake, 0.35% structural-deficit cap, modified by 2025 defense/infrastructure exemptions.
- UK: current budget targeted to be in surplus by 2029–30, plus debt-related rules.
- Most fiscal rules are framed as deficit, expenditure, or balance constraints.
- Debt anchors, where present, are often benchmarks rather than automatic state-contingent relaxation rules.
- Sustained austerity can make $\mathbb{E}_t \sum_j M_{t,t+j} \tau_{t+j}$ exceed what is compatible with P_t^* .

INTEREST-RATE NORMALIZATION CHANGES THE BACKING REQUIREMENT

- Post-GFC and COVID: public debt rose sharply across advanced economies.
- Normalization raises debt service for a given debt stock.
- CBO projects US net interest spending at 3.3% of GDP in 2026 and 4.6% in 2036.
- Those costs must be financed by higher taxes, lower spending, faster debt growth, default, or inflation.
- Higher debt raises the temptation to inflate $\Rightarrow$ time-inconsistency problem (Leeper–Leith–Liu).
- King (1995): disinflation can itself destabilize debt.

CONSOLIDATED GOVERNMENT BUDGET CONSTRAINT

A compact consolidated-government version is

$$\frac{B_{t-1} + H_{t-1} - A_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} M_{t,t+j} (\tau_{t+j} + \text{seigniorage}_{t+j}).$$

- Base money H_t and fiscal liabilities are jointly backed by fiscal resources, seigniorage, and assets.
- Open-market operations affect prices only if they change expected backing, liquidity services, risk, or constraints.
- Otherwise they are liability-composition swaps.

SUMMARY

MAIN RESULTS: REGIMES

- In the fixed-regime benchmark, determinacy requires **one unstable root** because there is one jump variable and one predetermined state.
- In **Regime M**, $g_M^\psi = 0$: active monetary / passive fiscal policy insulates inflation from fiscal shocks.
- In **Regime F**, $g_F^\psi \neq 0$: fiscal shocks move inflation because the price level revalues inherited nominal debt.
- Under Markov switching, equilibrium coefficients depend on the transition matrix $\mathcal{P}$, not only on today's policy coefficients.
- Reduced-form Taylor and surplus-debt regressions estimate **equilibrium comovements**, not structural regime parameters.

MAIN RESULTS: VALUATION

- Low government-debt returns do not by themselves determine FTPL uniqueness; the relevant object is the **risk-adjusted PV operator**.
- When uniqueness fails, fiscal valuation can still imply a **lower bound**, not a unique selector, for the price level.
- With defaultable debt, the valuation equation pins down **real repayment**, not the split between **inflation** and **default**.
- Inflation targeting requires **fiscal backing** consistent with the **target price path**.

APPENDIX

APPLICATION OF IDENTIFICATION CAVEAT: HISTORICAL REGIME LABELS

Period	Interpretation	Source
1942–1951	F (Fed pegged 3/8% T-bills)	Woodford (2001)
1960s–1979	F-like (weak Taylor response plus fiscal evidence)	CGG; Bianchi–Ilut
1980–2007	M-like, with switches	Chen–Leeper–Leith
2008–2015; 2020–2022	Ambiguous at ZLB	Open question
2022–2026	High-rate monetary tightening; fiscal side disputed	Open question
Brazil 2015	Loyo-style AM/AF interpretation	Leeper–Leith; Economist

- Regime labels require joint monetary–fiscal evidence, not a Taylor coefficient alone.

EMPIRICAL IMPLICATION: FISCAL-STRESS EXPECTATIONS CAN REVERSE INFLATION RESPONSES

- Stylized VAR fact: monetary easing is sometimes followed by **lower measured inflation**.
- Standard explanations include cost channels and omitted information.
- Switching explanation: in **Regime F** , or with probability of a switch to F , lower i_t today lowers expected future debt service.
- Lower expected **fiscal stress** can lower expected future inflation, hence lower current inflation.
- This is **one structural explanation** of price-puzzle responses, not a universal account.

THE 2008–2014 PUZZLE RECONSIDERED

- Federal debt rose sharply while the federal funds target was constrained near zero.
- Naive FTPL reading: higher nominal debt should imply much higher inflation.
- [Where is the inflation?](#)
- One reconciliation: lower discount rates, higher expected future backing, or stronger liquidity demand for safe debt raised the valuation of government liabilities.
- The price level need not rise one-for-one with nominal debt.
- Simple “debt up $\Rightarrow$ inflation up” is a wrong reading of the theory.

WALLACE IRRELEVANCE THEOREM

- At an interest-on-reserves Friedman rule ($R_t^M = R_t^B$), money and bonds are perfect substitutes.
- An open-market swap of money for bonds is then irrelevant:

$$\Delta M_t = -\Delta B_t \quad \Rightarrow \quad \text{no change in } P_t, c_t, R_t.$$

- Wallace (1981), Chamley–Polemarchakis (1984): Modigliani–Miller in monetary economies.
- Open-market operations matter when they change **fiscal backing**, **liquidity services**, **risk**, or constraints.

MODEL F VS MODEL E CENTRAL BANKS (SIMS 2004)

Model F (stylized Fed/BoE)

- CB holds mostly nominal public-sector debt.
- Positive seigniorage flows to Treasury.
- Balance sheet is essentially nominal.
- Needs Treasury fiscal backstop.

- Persistent negative net worth requires fiscal backing, future seigniorage, valuation gains, or institutional tolerance.

Model E (ECB, HKMA)

- CB holds real assets: gold, FX reserves.
- Fiscal backstop partly in-house.
- Natural for currency unions with heterogeneous fiscal authorities.
- FX reserves push the backing problem one level out.

SARGENT 1982: HYPERINFLATION ENDINGS AS FISCAL REGIME CHANGES

- Studied post-WWI hyperinflations in Austria, Hungary, Germany, Poland.
- All four ended abruptly after policy changes.
- Common pattern:
 - credible commitment to a balanced budget;
 - independence of the central bank from fiscal authority;
 - sometimes external conditionality.
- The historical interpretation is fiscal regime change, not monetary discipline alone.

BRAZIL 2015 AS A LOYO-STYLE AM/AF INTERPRETATION

- Constitutional benefit indexation and mandatory spending rules make roughly 90% of expenditures rigid or outside ordinary legislative control.
- 2015 overall public-sector deficit: about 10.3% of GDP; primary deficit: about 1.9% of GDP.
- Banco Central do Brasil raised policy rate to 14.25%.
- IPCA inflation reached 10.67% in 2015.
- Mechanism: higher i_t raises debt service; if fiscal policy cannot offset, expected inflationary/default adjustment rises.

BRAZIL 2016–2023: FISCAL RULE ADOPTION AND EROSION

- **December 2016:** Constitutional spending cap (*Teto de Gastos*) limited federal primary spending growth to previous-year inflation for 20 years.
- Interpretation: institutional commitment toward passive fiscal policy.
- Inflation: 10.7% (2015) → 2.9% (2017). Selic cut from 14.25% to 6.5% by 2018.
- **2020–:** COVID + political pressure effectively breach the cap.
- **2023:** Lula's government replaces the cap with a softer “Marco Fiscal Sustentável”.
- Lesson: fiscal-regime commitments can weaken over time.

ARGENTINA 2024: FISCAL CONSOLIDATION AS REGIME CHANGE

- Dec 2023: Milei takes office. Annual CPI inflation ~211%.
- Inherited: large primary deficits monetized by the BCRA.
- IMF description: 2024 fiscal adjustment around 5% of GDP.
- BCRA stops financing the Treasury; immediate currency devaluation.
- Monthly inflation: 25.5% (Dec 2023) → 2.7% (Dec 2024).
- 2024: primary surplus about 1.8% of GDP; overall cash surplus about 0.3% of GDP.
- Fiscal consolidation was central, but devaluation, recessionary compression, administered prices, and capital controls also mattered.

OPEN QUESTIONS

- How do we **identify the regime** in real time, before history settles it?
- How should **sovereign default risk** enter regime classification? (Open economy, monetary unions.)
- Can fiscal rules be redesigned to be **compatible with an inflation target**?
- What role for political economy in choosing which authority becomes **passive**?
- Does central-bank balance-sheet policy **blur the regime classification** entirely?